

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2022/2023

OCTOBER EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

FRIDAY, 21 OCTOBER 2022

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN SOFTWARE ENGINEERING

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions.

BACS2163 SOFTWARE ENGINEERING**Question 1**

You, as the project manager, have been assigned to lead a project team of 4 software engineers to develop a hotel reservation mobile application for DreamBay Hotel. The system should allow the customers to view rooms and promotion details, register as a member, make reservations and payments, feedback and review. The system should also allow the system administrator to manage users, rooms and reservations. After the initial requirements gathering process, you found that the requirements gathered were clear and could be modularized easily. Furthermore, the top and lower management of the client were very supportive to the upcoming project. Additionally, DreamBay Hotel expects the application to be simple, easy to use and all the transactions to be secured. Your team is given three (3) months to complete the application and to speed up the development process, your company has recently hired two part-time staffs.

- a) Identify if the hotel reservation mobile application as mentioned above is a generic software product or bespoke software product? Justify your answer with **TWO (2)** reasons.
(1 + 4 marks)
- b) Discuss **ONE (1)** reason why a *Plan-Driven Development* should be recommended for the above project.
(2 marks)
- c) Recommend **ONE (1)** software process model to develop the hotel reservation mobile application as mentioned above. Justify your recommendation with **FOUR (4)** reasons.
(2 + 8 marks)
- d) Explain **THREE (3)** software quality attributes for the hotel reservation mobile application as mentioned above. Justify your suggestions.
(6 marks)
- e) As a project manager, you would need to consider the factors that could motivate employees throughout the project. Identify and explain **ONE (1)** motivating factors for this case based on the Maslow Motivation Model.
(2 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 2**

Assume that you are a team leader in Zero IT Sdn. Bhd, a software development company in Penang. Your company has been offered a project to develop an online learning system for Fantastic Fashion Academy, a newly established institution in Penang which offers art courses. The courses are conducted in both full time and part time modes. Recently, the academy plans to switch to online platforms and has contacted your company to handle this project. You are responsible to build the online learning system with the core functions for the academy to allow customers to register as member, pay the fees, check classes' schedule, attend classes online and download materials in 3 months. Furthermore, the academy also wishes to extend their online system to share success story, subscribe to weekly newsletter, chat with company personnel, and connect to other social networks such as Facebook Page, Twitter, and Instagram in the next version.

Your team is given five (5) months to complete the system and to speed up the development process, your company has recently hired two part-time staffs. The project will start on December 2022 and from the initial meeting, you and your team has drafted main tasks that will be executed in the project along with its duration for the project schedule. The main tasks are requirements gathering, requirements analysis, data analysis, process design, data design, UI design, coding, testing and deployment, which all these main tasks must be performed sequentially. Requirements analysis will take about 2 weeks and data analysis will take about 3 weeks and both tasks can start immediately once the completion of requirements gathering task that expected to take 2 weeks. Once both analysis tasks ended, the process design, data analysis and UI design tasks will start one by one sequentially and all these design tasks are expected to take 2 weeks for each task. The coding tasks will take about 4 weeks and will start once the all the design tasks completed. The testing tasks will start once 50% of the coding task completed. And the testing task will take about 3 weeks and once it is completed the system will be deployed in a week time.

- a) Analyse **TWO (2)** techniques for discovering requirements for the above project. (4 marks)
- b) Identify **THREE (3)** functional requirements from the above online learning system and **TWO (2)** non-functional requirements. (5 marks)
- c) Prepare a Gantt Chart to develop the online learning system based on the information given. (9 marks)
- d) Assume that you are planning to adopt a viewpoint-oriented analysis method for the requirements analysis process. Identify **SIX (6)** viewpoints for the online learning system. Construct a viewpoint hierarchy to present your viewpoints. You must categorize your viewpoints under 3 categories with at least one viewpoint under each category. (7 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

WorldPrint Publication is a book publisher which started its first printed books in 1980s. The publisher is maintaining a file storage system since 1983 to store and manage the softcopy of books and books draft contents such as original story, story draft, supporting images or diagrams in their publication. The files might be useful in the future to trace the original source. The storage system has no documentation nor proper module design. The modules of the system have low cohesion and high coupling. The system has gone through a lot of modifications as the needs of their staff changing from time to time. However, since 2001 the vendor of the system has closed down their business and stopped providing maintenance to the system. Your company have been contacted by WorldPrint Publication to assist them in providing maintenance for their file storage system, however the publisher welcomes your suggestions.

- a) Based on your experience, discuss the **TWO (2)** factors that could affect the maintenance cost of WorldPrint's legacy system. (4 marks)
- b) Differentiate between *cohesion* and *coupling* in the above system design. Discuss how each of them would affect the maintainability of a system. (4 marks)
- c) Draw a diagram to explain legacy system assessment and management based on "Business Value" and "System Quality" areas. Explain which strategy you should apply to maintain WorldPrint's legacy system. (2 + 3 marks)
- d) Assume that WorldPrint IT department has decided to re-engineer the legacy system, propose **ONE (1)** system organization/structuring design model. Justify your answer. (6 marks)
- e) Suggest **THREE (3)** user interface design principles that must be applied in WorldPrint's proposed system. Justify your suggestions. (6 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

A small independent gym is planning to computerize a series of services such as gym booking and payment, facilities management, members management, and staff management. Users' membership cards will be converted into electronic identity (e-ID) to ease the access the series of services provided by the gym. The gym owner is also planning to implement automation of lighting during night in the washroom, air-con automated temperature detection and adjustment, and video monitoring system in the gym.

- a) Identify **TWO (2)** sets of stimuli-responses for the automation of a lighting system. (6 marks)

- b) Propose **TWO (2)** functional classification of Computer Assisted Software Engineering (CASE) tools with **TWO (2)** specific examples of tools for each classification that can be used by the gym. (8 marks)

- c) Plan the following testing techniques to test the gym booking module of the system. You must include **TWO (2)** examples of test cases for each of the testing technique.
 - (i) Black Box Testing (2 + 3.5 marks)
 - (ii) Stress Testing (2 + 3.5 marks)

[Total: 25 marks]